

Jessica SCHROUFF

Nationality: Belgium Residence: London, UK

EDUCATION

PhD in Electrical Engineering

Title: *Pattern Recognition in NeuroImaging: What can machine learning classifiers bring to the analysis of functional brain imaging?*

University of Liège, Belgium
Graduated March 13

Master's degree in Biomedical Engineering (M.Eng.)

Coursework included pattern recognition, digital signal processing, neuroimaging

University of Liège, Belgium
Graduated June 09

EMPLOYMENT

GlaxoSmithKline (GSK)

Director of Engineering, Responsible AI

- Developing and implementing Responsible AI best practices at GSK across applications and modalities (from supervised learning to LLM-based agents).
- Supervising multiple research projects, internally and with external collaborators.
- Managing a team of 13 members including managers, engineers and scientists from diverse backgrounds and levels of seniority.

London, United Kingdom
March/25 – present

Google DeepMind

Senior Research Scientist

- Leading major project on AI fairness across teams and product areas.
- Involved in projects referring to causality and large-scale models.
- Go-to person for fairness and interpretability, internally and externally.
- Manager for an intern and mentor for early career researchers.

London, United Kingdom
Nov/22 – March/25

Google Research / Health

Research Scientist, Senior Research Scientist

- Planification of research and experiments for the analysis of time series health care data using machine learning/deep learning methods.
- Leading major projects on model fairness (team of 50+), informativeness, and credibility.
- Initiated collaborations across product areas and geographical locations, leading to joint paper submissions.
- Manager for interns, AI resident and visiting faculty.
- Involved in the continuous learning of the team and mentorship of colleagues in terms of research.

London, United Kingdom
Jun/19 – Oct/22

Prodo.ai

Machine Learning Researcher

- Developed deep learning models for program source code analysis.
- Implemented state-of-the-art graphical deep learning models.
- In charge of disseminating scientific contributions (ICLR 2019 workshop).

London, United Kingdom
Nov/18 – May/19

Computer Science Department, University College London

Post-doctoral research fellow

- Developed new machine learning tools for multi-task learning and multiple kernel learning.
- Implemented machine learning modeling techniques for electrophysiological data.
- Designed a human declarative memory experiment for patients implanted with intracranial electrodes.
- Talked and chaired sessions at various conferences: OHBM (2017 and 2018), PRNI (Singapore, 2018), Brain Hack (San Francisco, 2017).
- Supervised students, from post-doctoral to undergraduate researchers.

London, United Kingdom
Jul/15 – Nov/18

Stanford University
Post-doctoral researcher

Stanford, CA USA
Sep/13 - Jun/17

- Developed a pre-processing pipeline for intracranial electrophysiological data in Matlab.
- Implemented multiple kernel learning for neuroimaging data.
- Talked at the Organization for Human Brain Mapping (OHBM 2014, 2015) annual meetings.
- Supervised Stanford undergraduate students.

University of Liège
PhD fellow

Liège, Belgium
Oct/09 - Sep/13

- Developed novel approaches to decode directed memory recall and resting wakefulness in humans.
- Implemented processing tools for electrophysiological sleep recordings in the 'fMRI Artefact rejection and Sleep Scoring Toolbox (FASST, www.montefiore.ulg.ac.be/~phillips/FASST.html) open-source software.

University Pierre et Marie Curie - INSERM
Intern

Paris, France
Feb/09 - May/09

- Investigated brain networks connectivity in resting wakefulness and anesthesia using unsupervised pattern recognition techniques.

SELECTED PUBLICATIONS ([Google Scholar](#))

- *Mind the Graph When Balancing Data for Fairness or Robustness.* J. Schrouff, A. Bellot, A. Rannen-Triki, A. Malek, I. Albuquerque, A. Gretton, A. D'Amour, S. Chiappa. 2024. [Advances in Neural Information Processing Systems](#).
- *Evaluating model bias requires characterizing its mistakes.* I. Albuquerque, J. Schrouff, D. Warde-Farley, A.T. Cemgil, S. Gowal, O. Wiles. 2024. [International Conference on Machine Learning](#).
- *Detecting shortcut learning for fair medical AI using shortcut testing.* A. Brown, N. Tomasev, J. Freyberg, Y. Liu, A. Karthikesalingam, J. Schrouff. 2023. [Nature Communications](#).
- *Adapting to Latent Subgroup Shifts via Concepts and Proxies.* I. Alabdulmohsin, N. Chiou, A. D'Amour, A. Gretton, S. Koyejo, M. Kusner, S. Pfohl, O. Salaudeen, J. Schrouff, K. Tsai. 2023. [International Conference on Artificial Intelligence and Statistics](#).
- *A reduction to binary approach for debiasing multiclass datasets.* I. Alabdulmohsin, J. Schrouff, S. Koyejo. 2022. [Advances in Neural Information Processing Systems](#).
- *Diagnosing failures of fairness transfer across distribution shift in real-world medical settings.* J. Schrouff, N. Harris, O. Koyejo, I. Alabdulmohsin, E. Schnider, K. Opsahl-Ong, A. Brown, S. Roy, D. Mincu, C. Chen, A. Dieng, Y. Liu, V. Natarajan, A. Karthikesalingam, K. Heller, S. Chiappa, A. D'Amour. 2022. [Advances in Neural Information Processing Systems](#).
- *Underspecification presents challenges for credibility in modern machine learning.* A. D'Amour, K. Heller, D. Moldovan, B. Adlam, B. Alipanahi, A. Beutel, C. Chen, J. Deaton, J. Eisenstein, M. D. Hoffman, F. Hormozdiari, N. Houlsby, S. Hou, G. Jerfel, A. Karthikesalingam, M. Lucic, Y. Ma, C. McLean, D. Mincu, A. Mitani, A. Montanari, Z. Nado, V. Natarajan, C. Nielson, T. F. Osborne, R. Raman, K. Ramasamy, R. Sayres, J. Schrouff, M. Seneviratne, S. Sequeira, H. Suresh, V. Veitch, M. Vladymyrov, X. Wang, K. Webster, S. Yadlowsky, T. Yun, X. Zhai, D. Sculley. 2022. [Journal of Machine Learning Research](#), [MIT Technology Review article](#).
- *Best of both worlds: local and global explanations with human-understandable concepts.* J. Schrouff, S. Baur, S. Hou, D. Mincu, E. Loreaux, R. Blanes, J. Wexler, A. Karthikesanligam, B. Kim. 2021. <https://arxiv.org/abs/2106.08641>.
- *Concept based model explanations for electronic health records.* D. Mincu, E. Loreaux, S. Hou, S. Baur, I. Protsuyk, M. Seneviratne, A. Mottram, N. Tomasev, A. Karthikesanligam, J. Schrouff. 2021. [ACM CHIL Proceedings](#).
- *PRoNT: pattern recognition for neuroimaging toolbox.* J. Schrouff, M.J. Rosa, J.M. Rondina, A.F. Marquand, C. Chu, J. Ashburner, C. Phillips, J. Richiardi, J. Mourão-Miranda. 2013. *Neuroinformatics* (11), 319-337.

PROFESSIONAL ACTIVITIES

Causality in Healthcare AI (CHAI) Hub
Industry Advisory Board member

UK
July/25 - present

Board of Women in Machine Learning

remote

Director (2020,21,23), VP of Programs (2022), Observer (2024-26)	Apr/20 - present
Workshop: Algorithmic Fairness through the Lens of ... Organizer, NeurIPS editions 2020-23, advisor NeurIPS 2024-25 and ICLR 2026, panelist at NeurIPS 2024	<i>virtual</i> Dec/20-24
NeurIPS conference Tutorial Chair	<i>virtual</i> Dec/22
Machine Learning for Healthcare (ML4H) symposium Area Chair (2020, 2021), Organizer (2021)	<i>virtual</i> Dec/21
Women in Neuroscience Repository (www.winrepo.org) Founder, project manager	<i>London, UK</i> Feb/18 - Mar/21
Pattern Recognition for Neuroimaging Toolbox, PRoNT Developer, technical support, supervised students, gave tutorials	<i>University College London</i> Aug/11 - Nov/18
Pattern Recognition for NeuroImaging workshop Organizer (2015), Member of steering committee	<i>Stanford University, USA</i> Jun/15 - Jun/18

FELLOWSHIPS AND AWARDS

- Award for exceptional contributions to diversity, equity and inclusion (Google UK), 2024
- Award for excellence in leadership (Google Research), 2022.
- Award for exceptional contributions to diversity, equity and inclusion (Google Research), 2022.
- Award for an exceptional contribution to site culture (Google UK), 2020, 2023.
- Marie Skłodowska-Curie Actions global fellowship (European Commission), 2015.
- Belgian American Educational Foundation (BAEF) fellow, 2013.
- Award from the Medical Rotary Foundation of Liege for a Translational Research Project (€15,000), 2013.
- F.R.I.A – National Scientific Research Funds (F.N.R.S. - Belgium) PhD fellowship, 2009.
- Erasmus internship fellowship, 2009.

ADDITIONAL INFORMATION

- **Area Chair:** ICLR 2025-26, TMLR (2022-2024), FAccT 2023, ML4H 2020, 2021.
- **Outstanding reviewer:** NeurIPS 2025 and 2020, ICLR 2020.
- **Reviewer** for NeurIPS, ICML, ICLR, ML4H workshop, ACM CHIL, FAccT, eLife, NeuroImage, Nature Medicine, PLoS, Science Advances and many more.
- **Programming:** Python, Matlab, basics in bash, HTML and CSS.
- **Tools:** Github, Latex, Visual Studio Code, Jupyter/Colab notebook, Gimp.
- **Languages:** English (fluent), French (native), Dutch (basics), Spanish (basics).
- **Interests:** Travels, Hiking, Advocating for Women in Science, Gymnastics, Reading, Dancing.